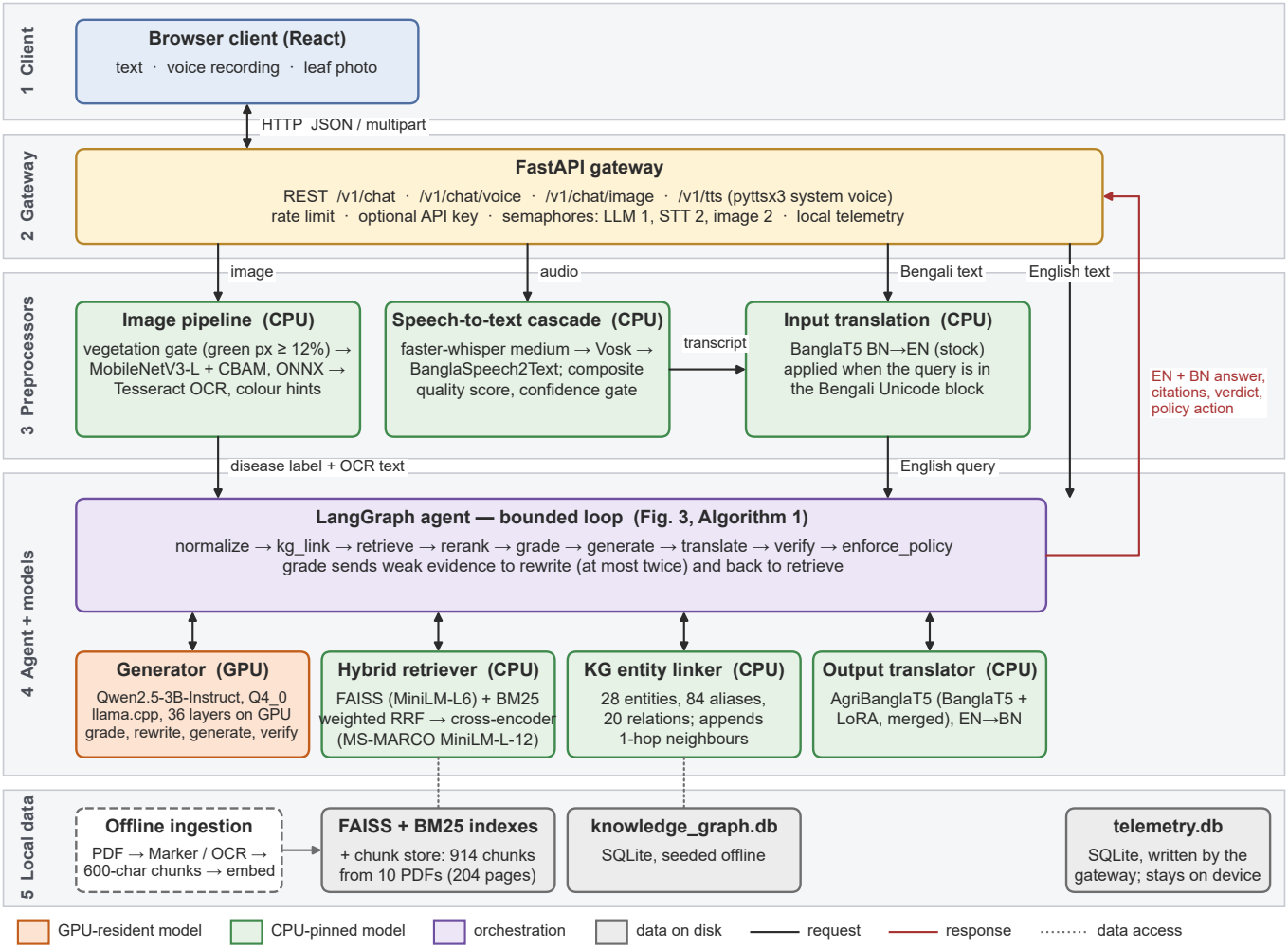

1 Client
Browser client (React)
text · voice recording · leaf photo
HTTP JSON / multipart
2 Gateway
FastAPI gateway
REST /v1/chat · /v1/chat/voice · /v1/chat/image · /v1/tts (pyttsx3 system voice)
rate limit · optional API key · semaphores: LLM 1, STT 2, image 2 · local telemetry
image
audio
Bengali text
English text
3 Preprocessors
Image pipeline (CPU)
vegetation gate (green px ≥ 12%) → MobileNetV3-L + CBAM, ONNX → Tesseract OCR, colour hints
Speech-to-text cascade (CPU)
faster-whisper medium → Vosk → BanglaSpeech2Text; composite quality score, confidence gate
transcript
Input translation (CPU)
BanglaT5 BN→EN (stock) applied when the query is in the Bengali Unicode block
EN + BN answer, citations, verdict, policy action
disease label + OCR text
English query
4 Agent + models
LangGraph agent — bounded loop (Fig. 3, Algorithm 1)
normalize → kg_link → retrieve → rerank → grade → generate → translate → verify → enforce_policy
grade sends weak evidence to rewrite (at most twice) and back to retrieve
Generator (GPU)
Qwen2.5-3B-Instruct, Q4_0 llama.cpp, 36 layers on GPU grade, rewrite, generate, verify
Hybrid retriever (CPU)
FAISS (MiniLM-L6) + BM25 weighted RRF → cross-encoder (MS-MARCO MiniLM-L-12)
KG entity linker (CPU)
28 entities, 84 aliases, 20 relations; appends 1-hop neighbours
Output translator (CPU)
AgriBanglaT5 (BanglaT5 + LoRA, merged), EN→BN
5 Local data
Offline ingestion
PDF → Marker / OCR → 600-char chunks → embed
FAISS + BM25 indexes
+ chunk store: 914 chunks from 10 PDFs (204 pages)
knowledge_graph.db
SQLite, seeded offline
telemetry.db
SQLite, written by the gateway; stays on device
GPU-resident model
CPU-pinned model
orchestration
data on disk
request
response
data access